SHAPE 8I: Bridging Benchmarks and Classrooms Through a Human-Centric Review of ELT-Bench

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Abstract

The contemporary language classroom is met with a dual challenge adapting to the ever-evolving digital landscape while also meeting diverse learners' needs. While an AI application can provide opportunities for personalized practice and quick assessment feedback, they often struggle with pedagogical competencies such as generating empathy, and adaptive guidance such as picking up behavioural clues. The variability in learners highlights the need for a human-centric teaching system that uses AI as an assisted tool. However, prominent AI tools are not evaluated against relevant benchmarks. To address this, our project evaluates these emerging benchmarks. We examine their structure and ability to capture the nuanced skills required in effective language teaching. By analysing how well the benchmarks represent real-life education needs and identifying areas for improvement, the study aims for the creation of, human-centric tools that guide responsible development and implementation of AI in language education.

1. Introduction

Benchmark evaluations play a vital role in measuring progress in AI. As these systems become increasingly embedded in education, their capabilities are generally judged across benchmarks developed for STEM-specific domains, such as mathematical reasoning or coding. However, ELT-Bench focuses on the pedagogical, interpersonal, and adaptive skills needed for effective language teaching. These are competencies that traditional benchmarks overlook.

Yet, even with ELT-specific benchmarks, some areas require refinement to better capture real classroom demands. Our analysis highlights the need down broad sub-competencies in course in course evalution to reduce variability, to incorporate holistic feedback practices that recognise learner strengths as well as areas from improvement, and to ensure performance evaluation tasks require explicit rubric setting so AI systems do not generate unsupported grading schemes. We also identify explainability as an overarching requirement. AI systems must clarify how they arrive at their conclusions, enabling more transparent and pedagogically grounded interactions.

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By refining these benchmarks, ELT-Bench can more effectively support, and equip language teachers with relevant technical insights, while allowing them to connect with learners on a human level.

This report evaluates ELT-Bench by analysing a sample of its tasks and scoring criteria to identify where the benchmark does not fully capture real classroom practice. We focus on issues in how competencies are defined, how tasks and rubrics are framed, and how criteria are applied, and conclude with targeted recommendations to strengthen validity and fairness. Detailed task examples are provided in the appendices.

2. Methodology

At the beginning, we met in person to clarify the overall mission and agree on our working approach, which involved allocating tasks according to page numbers in Dataset. Besides meeting online and offline, co-editing shared documents in the OneDrive folder proved to be an effective method. This allowed everyone to contribute their insights, and we also supported each other through the WhatsApp group.

During the six weeks, we completed 130 out of 356 tasks. In terms of the use of the tool, we cross referenced contents from the ELT-Bench tool to the Dataset. We began by locating the task ID in the Dataset first and then understanding the whole task requirements described in the Dataset, trying to imagine what an expected answer might contain. After that, we checked the criteria and compared the reference answer with the AI answer.

When it came to specific professional areas, we conducted additional research. This included learning more about CEFR level, how test validity is constructed, and exploring common concerns related to AI. As users, especially half of us are not native English speaker, we put ourselves in learner's and teacher's position to consider how AI can support us most effectively. As researchers, we consulted additional information to supplement areas outside our expertise.

3. Key findings

This section summarises the three key findings from our evaluation of ELT-Bench. We highlight the areas where refinements can better align the benchmarks with real classroom practices. These are tested against authentic cases provided in our team's dataset.

3.1 Competency Flaws

3.1.1 Overly Broad and Compound Sub-Competencies

To push towards ELT-Bench's aim to evaluate pedagogically meaningful behaviors, we have identified a few structural issues within the competency framework that limit the performance of the tool. The first flaw concerns **overly broad/compound sub-competencies,** especially those related to contextualisation. Many tasks require the model to adapt its response to learner variables such as age, proficiency, number of students, educational culture, purposes for study, and access to digital tools. However, these factors are combined into a singular sub-competency, creating ambiguity about what is being evaluated. Since task descriptions often lack sufficient contextual detail, the AI outputs become inconsistent, or the tool gives a response based on its assumptions of the learner (hallucination). For instance, tasks that rely on context appropriateness cannot be judged if the learner profile is only partially described, and in several cases, the reference answers themselves contradict or ignore missing contextual elements.

3.1.2 Feedback Competencies Omit Holistic, Strength-Based Feedback

The existing feedback sub-competencies emphasise identifying errors, prioritising areas to address, and offering explanations and practice activities. However, they don't explicitly require models to recognise and communicate a learner's strengths. This gap leads the AI system to provide feedback that is corrective, but not supportive, limiting its pedagogical usefulness and the effectiveness of the teaching-learning environment.

3.1.3 Performance Evaluation Lacks Rubric-Based Criteria

Tasks requiring the AI to assign a grade or overall evaluation often assume the presence of a rubric but don't include setting one as a sub-competency. Without explicit criteria, the tool has improvised grading schemes and applied inconsistent standards. The absence of a sub-competency requiring a rubric establishment or clarification reduces the transparency and reliability of the evaluation process. This also reduces the explainability of the system itself, because of it hallucinating the evaluation criteria.

See appendix A for further description along with Task Examples.

3.2 Framing Bias

Framing Bias refers to validity threats that arise from how task prompts and scoring criteria are presented. Even when tasks appear straightforward, the way they are framed can subtly direct, limit, or distort the model's response away from what is assessed. In ELT-Bench, this occurs when prompts are ambiguous, when essential contextual information is missing, or when cultural assumptions are implicitly encouraged by the rubric. These mechanisms can unintentionally "set the AI up for success or failure," influencing both the performance the model produces and the way its output is evaluated.

3.2.1 Ambiguity

Ambiguity occurs when task wording is unclear or underspecified, allowing multiple reasonable interpretations even though the benchmark expects only one. This threatens validity because responses aligned with plausible readings of the prompt may be downscored if they do not match the rubric's unstated expectations. In ELT-Bench, ambiguity often results from misalignment between the prompt and the competency criteria. This criteria-expectation may come from reference-answer default bias (see 3.3.1). For example, some prompts never specify "key language" or require no teaching framework, while rubrics reward them. Ambiguous evaluative terms like "sophisticated" also create interpretation gaps for both AI and assessors, resulting in inconsistent output and inconsistent scoring. See Appendix B1.

3.2.2 Information Insufficiency

Information insufficiency arises when prompts omit essential contextual details, including class size, institutional type, learner goals, or cultural norms, when the scoring criteria expect the AI to tailor its response to these factors. This is the task-prompt side of broad-competency flaw (3.1.1).

Because the benchmark often provides only partial profiles (typically just age, level, location), the model must guess unstated pedagogic or cultural constraints, and assessors must decide whether to penalise or overlook this. This weakens validity by rewarding successful guesswork rather than genuine ability to adapt to given contexts. It is especially pronounced in tasks involving culturally sensitive communication, where unprovided information forces the AI to rely on its own assumptions. See Appendix B2.

3.2.3 Bias

Bias occurs when prompts or rubrics embed cultural values or assumptions that implicitly define what counts as an "appropriate" answer, thereby framing the AI's output toward particular worldviews. This happens when scoring criteria expect local cultural, economic, or behavioural insights that the prompt never requests, pushing the model to fill gaps with potentially stereotyped or culturally loaded assumptions. In some items, even the reference answers adopt generalisations (e.g., linking English proficiency to "better jobs" or attributing personality traits to nationalities), signalling to models and assessors what kinds of cultural narratives are rewarded. Such framing penalises prompt-faithful responses while privileging speculative or normatively aligned ones. See Appendix B3.

3.3 Criteria Flaws

Regarding the criteria, the accuracy of ELT-benchmark may be affected by shortcomings in it. Although the benchmark is comprehensive, these flaws could generate unreliable scores and rubrics, potentially compromising the validity of the whole benchmark. These flaws in the criteria can be categorized into three main sub-groups.

3.3.1 Default bias

Default bias refers to the researchers presetting the criteria while neglecting the real task context. In other words, some criteria seem to be established based on the reference answer, leading to a mismatch between criteria and AI answer, which is constrained by insufficient information. This is interconnected with framing bias and is *supported by the examples of tasks (task_ID = 5, task_ID = 6) discussed in appendix B1.*

3.3.2 Inconsistency

Some tasks that are designed to evaluate similar competencies have different marking criteria, resulting in uncertainty within the benchmark.

In terms of the score and rubric wording inconsistency, some tasks share identical requirements, but their criteria are similar without being exactly the same, leading to increased confusion. A similar inconsistency appears in the focus areas; this can be illustrated briefly by task_ID = 282, and 288. Both tasks are about designing the course and having similar requirements. Nevertheless, they have different criteria. Is the same skill considered important for the first task but not for the other one? Both factors carry equal weight.

Overlooking these inconsistencies in the criteria, whether in wording or evaluative focus, contributes to a lack of trust in the benchmark.

3.3.3 Incompleteness

Moving on to the content of the criteria, several components appear not to have been thoroughly considered across categories, resulting in superficial treatment and fragmented information. A notable example of such incompleteness can be observed in task_ID = 93. The intention of evaluating specific ability can be seen, but the criteria is too simplified to assess it appropriately. Furthermore, certain criteria adopt only one side perspective, overlooking essential complementary aspects, as seen in task_ID = 160, and 178. It remains significant for students to understand both right and wrong within the learning process. However, the "giving feedback" competency does not account for the importance of highlighting mistakes.

Across all tasks, two universal criteria consistently lead to score reduction. Given that the tool may be accessed by learners directly, safety considerations should be given greater emphasis and assigned higher weighting.

Overall, the lack of completeness in the criteria compromises the intended assessment objective. As a result, the expected outcomes may not be achieved, thereby weakening the validity of the benchmark.

See appendix C for further description along with Task Examples.

4. Recommendations

This section presents targeted revisions that correspond to our key findings above. Each solution is stated succinctly and tied to practical changes in the competency framework.

4.1 Competency Flaws

4.1.1 Reduce Learner Variability by Splitting Overly Broad Sub-Competencies

Revise Course Planning sub-competency ‘a’ into discrete, testable components:

- a1. Identify learners' aims (purpose for learning: academic, professional, and personal).
- a2. Identify learner needs (proficiency gaps, accessibility).
- a3. Consider contextual constraints (class size, instituition type, access to technology).
- a4. Integrate learner interests (topics, genres, and personal motivations).

This decomposition helps personalize course structure in terms of designing course materials, assessments, evaluation criteria, and feedback structure to each student.

4.1.2 Require Holistic, Strength-Based Feedback

Add an explicit feedback sub-competency: **e. Identify and communicate learner strengths. (weight = 7).**

Update scoring rubrics for the existing items (a-d) to require balanced feedback:

- Error diagnosis + at least one strength or success statement.
- A self-evaluation prompt.

This encourages the tool to produce formative, motivational feedback that mirrors effective teacher practice.

4.1.3 Anchor Performance Evaluation with Rubrics and Justification

Add a dedicated performance-evaluation sub-competency: **b. Establish or request explicit evaluation criteria (rubric) before assigning a grade or level. (weight = 8)**

Require models to

- Present the rubric they used, or
- Explicitly request one if none is provided.

Scoring should reward transparent justification (mapping mrks to rubric descriptors) and penalise unsupported numeric or CEFR assignment. This reduced hallucinated grading and increases interpretability.

4.2 Framing Bias

- Align prompts and rubrics: Ensure every scored element appears in the task instructions, and that no rubric requirement depends on hidden assumptions. Specific ways to do it:

    - Make target constructs explicit: If the rubric expects “key language” or specific lesson-planning frameworks, these should be stated in the prompt, so reasonable alternative interpretations are not penalised.
    - Provide essential contextual information: When adaptation to context is scored, prompts should specify factors such as class size, institutional setting, digital access, or cultural norms; if unavailable, they should not be assessed.
    - Revise universal criteria: Revise UC2, so it evaluates only the contextual elements actually present in the prompt.
- Define evaluative terms operationally: Replace vague terms like “sophisticated” or “appropriate” with clear linguistic or pedagogic descriptors (e.g., CEFR features, communicative outcomes).
- Do not rely on unstated cultural knowledge: Tasks requiring culturally specific behaviour should supply the relevant norms directly or provide them through a system prompt.
- Remove culturally loaded statements: Rubrics and reference answers should avoid unneeded cultural hierarchies or generalisations unless central to the construct.

4.3 Criteria Flaws

Even when criteria are applied across different tasks, those assessing the same competency should employ consistent wording to avoid ambiguity. Similarly, despite tasks may target various aspects within the same broader category, any skill that is relevant across tasks should be accompanied by aligned and consistent assessment criteria.
Besides consistency, the criteria should be reviewed for each specific skill within a competency to find any weaknesses or gaps. Ensuring both the consistency and completeness of the whole criteria not only promotes fairness but also enhances the comprehensiveness and reliability of the evaluation of outcomes.

5. Conclusion

This report set out to identify general patterns of flaws within ELT-Bench for refinement by systematically analysing a given sample of its tasks, criteria, and reference answers. Through cross-checking 130 task descriptions with scoring expectations and AI/reference outputs, we found recurring issues across three areas: competency flaws, framing biases, and criteria of inconsistencies.

Competency flaws arose from overly broad sub-competencies, the absence of strength-based feedback requirements, and performance evaluations that lacked rubric grounding. Framing biases emerged from ambiguous prompts, missing contextual information, and culturally loaded assumptions that shaped how responses were judged. Criterial flaws included default bias toward reference answers, inconsistent wording and emphasis across similar tasks, and incomplete treatment of key pedagogical skills.

To address these issues, we recommend separating compound sub-competencies, requiring balanced feedback, grounding evaluations in explicit rubrics, aligning prompts with criteria, defining evaluative terms operationally, avoiding unstated cultural assumptions, and ensuring consistent and complete criteria across tasks. These targeted revisions directly respond to the patterns identified and strengthen the validity, fairness, and interpretability of ELT-Bench.

6. References

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Appendix

A. Key Findings on Competency Flaws

A1. Overly Broad and Compound Sub-Competencies

Some sub-competencies, for example, under “Course Planning” are compounded in nature, making it unclear which dimension isbeing assessed. The most notable example is:

a. **Decide which learning goals are most important for the students’ learning aims, their context, their needs and interests.**

This sub-competency merges four pedagogical factors, resulting in ambiguity which can be bridged by separating them.

Evidence from Tasks:

- Many tasks supply only one or two learner variables (e.g., age and level), leaving gaps around purpose, interests, or contextual constraints.
- Scorers must infer which missing elements matter, causing unpredictable variation.
- In tasks such as task_ID = 106, 210, reference answers incorporate contextual details absent from the prompt, demonstrating inconsistency in how context is interpreted.

Consequences for Benchmark Reliability:

- AI outputs may appear incomplete due to incomplete task specifications.
- Different course structures emphasise different contextual dimensions.
- Benchmark difficulty is influenced more by task-writing variability than by AI capability.

A2. Feedback Competencies Omit Holistic, Strength-Based Feedback

The current sub-competencies for “Giving Feedback” (a-d) focus entirely on identifying errors, prioritising them, offering explanations, and providing improvement activities. None require communication of what the learner has done well.

Evidence from Tasks:

- In task_ID = 160, the model provided exclusively corrective feedback, while the reference answer highlighted strengths in organisation and clarity.
- Across tasks, models display a tendency toward deficit-framed feedback because the competency requires only error-based reasoning.

Consequences for Benchmark Reliability:

- AI appears less pedagogically aligned with real-world teacher practice than it may actually be

- Learners would receive feedback that is accurate but demotivating or incomplete.
- Tasks do not assess an essential component of formative assessment.

A3. Performance Evaluation Lacks Rubric-Based Criteria

The sub-competency under “Performance Evaluation”:

a. **Assign an evaluation of the performance as required**

This sub-competency does not specify how evaluations should be grounded (e.g, marking criteria, CEFR descriptors, task requirements).

Evidence from Tasks:

- In tasks without rubrics (e.g., task_ID = 16), models sometimes invent grading schemes or apply mismatched criteria.
- When rubrics are provided (e.g., task_ID = 178), AI responses are significantly more consistent and transparent.
- Differences in age, proficiency, or task type influence evaluations unpredictably due to missing guidance on what standards to apply.

Consequences for Benchmark Reliability

- Scores may reflect the model’s improvisation rather that its pedagogical reasoning.
- Evaluations lack transparency, making it difficult to assess the quality of the tool’s judgement.
- Scoring inconsistency increases when tasks impllicitlu expect a rubric, but do not supply one.

B. Key findings on Framing Bias

B1. Ambiguity

Core issue: Prompts allow multiple valid interpretations, but rubrics reward only one. This misalignment penalises plausible responses and creates scoring inconsistency.

Examples:

Undefined linguistic expectations (task_ID = 207)

- Prompt: “Have a conversation as a friendly classmate.”
- Rubric: Rewards use of unspecified “key language”.
- Flaw: The model cannot know which forms qualify as “key language,” so natural conversational responses may be unfairly down-scored.

Unsignalled methodological expectations (tasks_ID = 5 & 6)

- Prompt: Produce a lesson plan, no framework specified.
- Rubric: Rewards recognised frameworks (PPP, ESA, TBLT).
- Flaw: Multiple sequencing choices are legitimate, but only those matching hidden rubrical preferences score well.

Undefined evaluative terms (task_ID = 288)

- Rubric: Requires “sophisticated diplomatic English.”
- Flaw: “Sophisticated” could mean advanced lexis, pragmatic nuance, intercultural tact, or structural complexity.
- Impact:
    - Models may generate different interpretations of sophistication.
    - Human assessors may apply the term differently, weakening reliability.

Why this matters: Ambiguity creates variance in both model output and human judgement, reducing construct alignment.

B2. Information Insufficiency

Core issue: Tasks omit essential contextual details, yet rubrics expect adaptation to context. This pushes the model to guess and forces assessors to interpret how much inference is acceptable.

Examples:

Underspecified learner profile (task_ID = 288)

- Prompt: Design training for diplomatic English, but no indication of whether the context is local (e.g., Kosovo-specific) or international.
- Flaw: Course content, register, and scenarios differ radically across contexts, yet the model must infer the unstated one.

Universal Criterion 2 requires adaptation to missing details

- Criterion: Considers age, class size, institutional culture, digital access, etc.
- Reality: Most tasks provide only age, level, and country.
- Flaw: AI cannot demonstrate adaptation to variables it was not given; assessors vary in whether they ignore or expect inference.

Culturally dependent behaviour tasks (task_ID = 308)

- Prompt: Does not state cultural norms or display rules needed for “appropriate” behaviour.
- Flaw: Model must supply assumptions about cultural expectations, creating inconsistent or biased outputs across models.

Why this matters: The benchmark ends up rewarding the quality of the model’s guesses rather than its ability to respond to provided context.

B3. Bias

Core issue: Prompts or rubrics encode cultural assumptions that implicitly define what counts as “appropriate,” favouring particular worldviews and penalising culturally neutral responses.

Examples:

Rubric requiring culturally specific economic knowledge (task_ID = 33)

- Prompt: Set learning goals for A2 adults in Kenya seeking better employment opportunities.
- Rubric: Requires consideration of “local employment sectors in Kenya.”
- Flaw: The prompt never asks for economic sector analysis; meeting the criterion requires speculative or stereotyped assumptions about Kenya’s labour market.

Culturally loaded reference answers (task_ID = 347)

- Statements include:
  - “Students who speak English well get better jobs.”
  - “Nepali students, who are generally hardworking, often study abroad.”
- Flaws:
  - Implies English is inherently superior or tied to economic mobility.
  - Frames overseas study as normative or aspirational.
  - Makes generalisations about national character (“hardworking”).
- Impact: Suggests to models and assessors that culturally specific value judgments are desirable.

Why this matters: Rubrics and reference answers can implicitly reward culturally narrow or stereotyped viewpoints, disadvantaging neutral or context-faithful responses.

C. Key findings on Criteria Flaws

C2. Inconsistency

a. Wording Inconsistency

Take task_ID = 277, 280 and 281 as an example. These three tasks fall under category course planning and assess planning learning goals ability. The task criteria beneath them are essentially intended to addressing the same issue or at least they should. (Although one criteria mentioned considering cultural context but others do not, cultural context is equally important across all these tasks.) However, they use different descriptions and scores which leads to confusion.

- task_ID = 277, task criteria 04 “Considers cultural and educational context of Astana, Kazakhstan \\8”
- task_ID = 280, task criteria 05 “Ensures goals are appropriate for university students aged 18-22 in Ashgabat \\8”
- task_ID = 281 shows the same issue but divides into two task criteria
  “06. Considers the Georgian educational context for young learners \\7”
  “04. Ensures all goals are age-appropriate and achievable for 7-9 year olds \\9”

b. Emphases Inconsistency

task_ID = 282 and 288 are about designing the course, but they have different criteria.

- task_ID = 282, consensus criteria 01b-01 “learning goals progress in difficulty over the course, and complementary goals are grouped together into units and lessons \\ 4
- task_ID = 288, consensus criteria 01c-01 “Pace, recycling, assessment/progress checking, amount of practice, balance on online/in-person, use of digital tools, etc, are appropriate for the teaching context and learner profile \\8.”

Nevertheless, 01c-01 is also important in task_ID = 282 in fact.

C3. Incompleteness

a. Oversimplify

Although the task and prompt are mentioned for scaffolding and are intended to assess it, the criteria remain overly simplified and could not assess it appropriately.

- task_ID = 93, assessing sub-competency 2 “b. Provide appropriate level of scaffolding” using Consensus criteria “03b-01. Activity includes scaffolding at earlier stages of learning and reduces it as learning progresses \\6”, While it appears to focus on scaffolding, it fails to articulate the importance of appropriately removing scaffolding as the learner gains a better understanding of the material.

b. Score Reduction

There are two universal criterions that appear beneath every task. However, given that as an educational tool, the AI model should not produce any answer that needs to be approved by the teacher in case of it upsetting people, universal criteria 05 should be taken more seriously, potentially through adjusted the score.

- task_ID = 93, universal criterion 04 “The response includes content that is likely to offend, upset or shock the learners or teachers it is aimed at \\\\-10”|
- universal criterion 05 “The response includes content that needs the teacher to provide sensitive guidance and judgement because it might upset some people in the class \\\\-5”